Summary of Changes

Management and Analysis of Short Texts: A Graph-Based Approach

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This document summarises the changes made to the dissertation after the internal defence in response to the remarks of the jury members, together with several other improvements to the text and print quality. Locations refer to the revised version and identify the section and paragraph, e.g., 1.2.2 ¶ 4 denotes the fourth paragraph of Section 1.2.2. Unnumbered subsections are indicated by their title.

General

Line numbers	Removed the proof line numbers.
Language	Corrected spelling and writing errors throughout the dissertation and improved the phrasing of several sentences, following the suggestions of the jury members (too numerous to list individually).
Terminology	Standardised terms and wording throughout the dissertation to improve consistency and readability (too numerous to list individually).
Images	Replaced several images with higher-resolution versions for better readability and print quality.
Page size	Changed the page size to 160×240 mm, following the faculty guidelines, and adjusted the margins accordingly.
Layout	Made further layout adjustments, such as repositioning and resizing some objects to improve the overall appearance and readability for the printed version.
Bibliography	Checked that the URLs in the bibliography are still valid and updated the “visited on” dates accordingly.
Bibliography	Updated the formatting of the bibliography to ensure consistency over the entire document and a proper citation style.

Preliminary Pages

Dankwoord	**Acknowledgements.** Wrote the acknowledgements.
Summary	**Accessibility.** Revised the summary to make it more accessible and readable while retaining its technical nature. The content is unchanged, but the language has been improved, some sentences have been reworded, and some clarifications have been added (too numerous to list individually).
Samenvatting	**Dutch summary.** Made the same revisions to the Dutch summary to match the updated English version, and adjusted some translations to make the Dutch more accurate and natural (too numerous to list individually).

Chapter 1: Introduction

1.2.2 ¶ 4	**Interpretability.** Softened the claim that more complex models are generally less interpretable, acknowledging that this is not always the case and that the claim mainly refers to deep neural networks as black boxes.
1.2.3 ¶ 5	**Foundation models.** Mentioned foundation models, which can (partially) generalise across languages but may struggle with low-resource languages, depending on the transferable features of the language and the availability of training data.
1.3.1 ¶ 2–3	**Epigram types.** Made small additions to the introduction of the epigram types to emphasise that they represent one defensible scholarly interpretation of the corpus rather than an objective ground truth, in line with the later chapters.
1.3.2 ¶ 2 and 1.4.2 ¶ 2	**Embeddings and NLP tools.** Slightly softened the claim that embedding approaches and other NLP tools are entirely unsuitable for Byzantine texts. Acknowledged that some exist, but clarified that these are mostly based on normalised or edited texts, which is not the case for the DBBE corpus.
1.3.3 ¶ 1	**Intertextuality.** Explicitly mentioned the problem of intertextuality, explained what it entails, and noted that this work can contribute to it.
1.6 item 3	**Publication details.** Added the publication details (DOI, volume, ...) of *Verse, Variation, and Meaning: Annotating Semantic Distance in Greek Marginal Texts*, which has since been published.
1.6 item 7	**Publication details.** Added the publication details (DOI, volume, ...) of *Ranking L-grades for Veracity Handling in Logic Reasoning*, which has since been published.

Chapter 2: A Graph Data Model for Short Interconnected Texts

Opening	**Epigraph.** Replaced the epigraph by Abbie Hoffman, which did not really fit the thesis, with a quote by George E. P. Box.
2.3.1 ¶ 1	**Indexing.** Made the indexing in this section consistent: the characters of a text and its substrings are now both indexed from 0, whereas previously one was indexed from 1 and the other from 0.
2.6.1 ¶ 2 and 2.7 ¶ 4	**Spelling shifts.** Added a short discussion of other shifts in the Greek language beyond itacism, to clarify how handling them would affect the graph size and to consider them for future work.

Chapter 3: A Hierarchical Orthographic Similarity Measure for Short Texts

3.5.1 ¶ 2	**Expert groupings.** Clarified the limitations of the expert-curated groupings, in line with the discussions in later chapters: they represent one defensible scholarly interpretation rather than a perfect ground truth.
3.5.2 ¶ 3 *Two-Level String Matcher*	**Parameter α_Q.** Clarified that the parameter α_Q of the Two-Level String Matcher (denoted simply as α in the earlier version of the thesis) differs from the threshold α of the α-cut in the validation process.
3.5.4 Figure 3.9 *Query-Based Grouping of Similar Text Nodes*	**Figure labels.** Fixed the epigram ID labels in Figure 3.9, which were swapped between epigrams 22826 and 21182, and corrected the subsequent discussion accordingly, including the groups and word positions of these two epigrams. This should resolve the confusion about the results in this figure raised in one of the reading reports.

Chapter 4: A Context-Aware and Responsible Similarity Measure for Short Texts

Opening	**Epigraph.** Changed the attribution of the epigraph to indicate that it is a popular paraphrase of a quote by Aristotle rather than a direct quote.
4.4.2 Figure 4.1	**Figure caption.** Extended the caption to explain the aggregated similarity s and the confidence score c shown in the figure.
4.5.4 ¶ 2, 5 *Quality Measure Results*	**Precision and recall.** Added a more fine-grained discussion of the different trends in precision and recall in the experiments of this chapter, such as how the metadata raise recall across most thresholds, so that the focus no longer lies solely on the F_1-score.

Chapter 5: Ranking of Satisfaction-Confidence Pairs

Opening	**Publication details.** Added the publication details (DOI, volume, ...) of *Ranking L-grades for Veracity Handling in Logic Reasoning*, which has since been published.
5.3.2 ¶ 5	**Injective.** Briefly explained the term "injective" where it is used in the context of preorders, as no definition was provided.
5.4.1 ¶ 1 and 5.4.2 ¶ 2	**Tie relation.** Introduced a separate tie relation $\sim$ for equally preferred L-grades, instead of using equality with sub- and superscripts as this could be confused with the standard equality between L-grades.

Chapter 6: Context-Aware and Responsible Short Text Clustering

6.5.2 ¶ 3 *Similarity Inputs and Weight Vector Configurations*	**Weight vector notation.** Clarified the notation of the optimised weight vectors and added an example to avoid confusion.
6.5.2 ¶ 3–4 *Quality Measures, Adjusted Mutual Information (AMI)*	**ARI and AMI.** Clarified the definitions of, and differences between, the Adjusted Rand Index (ARI) and the Adjusted Mutual Information (AMI). Also added a small example to illustrate these differences and explained how the two measures relate when one is higher or lower than the other.
6.5.3 ¶ 3 *Context-Aware Similarity Clustering*	**Precision and recall.** Added a more fine-grained discussion of the different trends in precision and recall, rather than focusing mainly on the F_1-score: the metadata raise recall and lower precision for all three algorithms.

Chapter 7: Conclusions and Future Work

7.2.2 ¶2	**Embeddings and NLP tools.** Slightly softened the claim that embedding approaches and similar techniques are entirely unsuitable for Byzantine texts.
7.2.3 ¶5	**Precision and recall.** Carried the more fine-grained discussion of precision and recall from Chapters 4 and 6 over to the conclusions, rather than focusing mainly on the F_1-score.
7.5 ¶2	**Spelling shifts.** Added brief remarks on the potential inclusion of other spelling shifts beyond itacism in future work, and on the potential advantages of doing so.
7.5 ¶2	**Embeddings and NLP tools.** Outlined future work to incorporate embedding-based semantic similarities into the proposed methods as additional similarity inputs.
7.5 ¶3	**Outlier detection.** Added ideas on outlier detection as future work, e.g., to identify epigrams that fail to connect to the rest of their expert-curated group.
7.5 ¶3	**PageRank.** Mentioned further analysis techniques for future work, including but not limited to PageRank.
7.5 ¶4	**Generalisation and complexity.** Made it even more explicit that experiments on other corpora are needed to validate the generalisability of the proposed methods, and described how much larger corpora would affect the computational complexity of the current approach.
7.5 ¶4	**Genre and register.** Mentioned the need to analyse the genre and register of the texts in future work, to better understand how these factors affect the proposed methods.
7.5 ¶4	**Intertextuality.** Mentioned the potential of these methods for the broader problem of intertextuality.